

# UNIT-I: LOGIC AND PROOF TECHNIQUES

## 2. CONDITIONAL STATEMENTS AND LOGICAL EQUIVALENCE

### 2.1 Conditional Statements (Implication)

One of the most important logical connectives is the conditional statement or implication.

A conditional statement has the form

$$p \rightarrow q$$

and is read as:

"If p, then q."

where

- p is called the hypothesis, antecedent, or premise.
- q is called the conclusion, consequent, or result.

A conditional statement describes a relationship between two propositions.

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### Examples of Conditional Statements

#### Example 1

If a number is divisible by 4, then it is even.

Here

p : The number is divisible by 4.

q : The number is even.

Since every multiple of 4 is even, the implication is true.

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## Example 2

If it rains, then the roads become wet.

$p$  : It rains.

$q$  : Roads become wet.

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## Example 3

If a student studies regularly, then the student performs well in examinations.

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## Truth Table of Implication

| $p$ | $q$ | $p \rightarrow q$ |
|-----|-----|-------------------|
| T   | T   | T                 |
| T   | F   | F                 |
| F   | T   | T                 |
| F   | F   | T                 |

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## Understanding the Truth Table

### Case 1

$p = \text{True}$

$q = \text{True}$

Example:

If a number is divisible by 4, then it is even.

8 is divisible by 4 and 8 is even.

Therefore the implication is true.

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## Case 2

$p = \text{True}$

$q = \text{False}$

Example:

If a number is divisible by 4, then it is odd.

8 is divisible by 4 but not odd.

Therefore the implication is false.

This is the only situation in which an implication is false.

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## Case 3

$p = \text{False}$

$q = \text{True}$

Example:

If 5 is divisible by 4, then 5 is prime.

Since 5 is not divisible by 4, the hypothesis is false.

Therefore the implication is considered true.

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## Case 4

$p = \text{False}$

$q = \text{False}$

Example:

If 5 is divisible by 4, then 5 is even.

Again the hypothesis is false.

Therefore the implication is considered true.

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## Important Examination Note

A conditional statement is false only when:

The hypothesis is true and the conclusion is false.

This fact is frequently used in proofs and truth-table problems.

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## 2.2 Converse of a Conditional Statement

The converse of

$$p \rightarrow q$$

is

$$q \rightarrow p$$

The converse is obtained by interchanging the hypothesis and conclusion.

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### Example

Original Statement:

If a number is divisible by 4, then it is even.

Converse:

If a number is even, then it is divisible by 4.

This statement is false.

Counterexample:

6 is even but not divisible by 4.

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### Important Observation

A statement and its converse are generally not logically equivalent.

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## 2.3 Inverse of a Conditional Statement

The inverse of

$$p \rightarrow q$$

is

$$\neg p \rightarrow \neg q$$

It is obtained by negating both the hypothesis and conclusion.

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### Example

Original Statement:

If a number is divisible by 4, then it is even.

Inverse:

If a number is not divisible by 4, then it is not even.

False.

Counterexample:

6 is not divisible by 4 but is even.

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## 2.4 Contrapositive

The contrapositive of

$$p \rightarrow q$$

is

$$\neg q \rightarrow \neg p$$

The hypothesis and conclusion are interchanged and negated.

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## Example

Original Statement:

If a number is divisible by 4, then it is even.

Contrapositive:

If a number is not even, then it is not divisible by 4.

This statement is true.

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## Important Theorem

A conditional statement and its contrapositive are logically equivalent.

That is,

$p \rightarrow q$

and

$\neg q \rightarrow \neg p$

always have identical truth values.

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## Summary Table

| Statement Type | Form                        |
|----------------|-----------------------------|
| Conditional    | $p \rightarrow q$           |
| Converse       | $q \rightarrow p$           |
| Inverse        | $\neg p \rightarrow \neg q$ |
| Contrapositive | $\neg q \rightarrow \neg p$ |

Important:

- Conditional and Contrapositive are logically equivalent.
  - Converse and Inverse are logically equivalent.
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## 2.5 Biconditional Statements

A biconditional statement has the form

$$p \leftrightarrow q$$

and is read as

"p if and only if q."

or

"p iff q."

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### Meaning of Biconditional

The statement

$$p \leftrightarrow q$$

means both

$$p \rightarrow q$$

and

$$q \rightarrow p$$

are true.

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### Example

An integer  $n$  is even if and only if  $n = 2k$  for some integer  $k$ .

This statement has two parts:

Part 1:

If  $n$  is even, then  $n = 2k$ .

Part 2:

If  $n = 2k$ , then  $n$  is even.

Since both are true, the biconditional is true.

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## Truth Table

| $p$ | $q$ | $p \leftrightarrow q$ |
|-----|-----|-----------------------|
| T   | T   | T                     |
| T   | F   | F                     |
| F   | T   | F                     |
| F   | F   | T                     |

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## 2.6 Necessary and Sufficient Conditions

These concepts frequently appear in theorem statements.

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### Sufficient Condition

If

$$p \rightarrow q$$

then  $p$  is a sufficient condition for  $q$ .

Example:

Being divisible by 4 is sufficient for being even.

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### Necessary Condition

If

$$p \rightarrow q$$

then  $q$  is a necessary condition for  $p$ .

Example:

Being even is necessary for being divisible by 4.

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## Example

"If a student gets admission to a PhD program, then the student has a master's degree."

Master's degree is necessary.

Admission to PhD is sufficient.

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## 2.7 Logical Equivalence

### Definition

Two compound propositions are logically equivalent if they have identical truth values under every possible assignment of truth values.

Notation:

$$p \equiv q$$

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### Example

Show that

$$p \rightarrow q$$

is logically equivalent to

$$\neg p \vee q$$

Truth Table:

| p | q | $p \rightarrow q$ | $\neg p \vee q$ |
|---|---|-------------------|-----------------|
| T | T | T                 | T               |
| T | F | F                 | F               |

| p | q | $p \rightarrow q$ | $\neg p \vee q$ |
|---|---|-------------------|-----------------|
| F | T | T                 | T               |
| F | F | T                 | T               |

Since the final columns are identical,

$$p \rightarrow q \equiv \neg p \vee q$$


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## 2.8 Tautologies, Contradictions, and Contingencies

### Tautology

A proposition that is always true.

Example:

$$p \vee \neg p$$

Truth Table:

| p | $p \vee \neg p$ |
|---|-----------------|
| T | T               |
| F | T               |

This is called the Law of Excluded Middle.

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### Contradiction

A proposition that is always false.

Example:

$$p \wedge \neg p$$

Truth Table:

| p | $p \wedge \neg p$ |
|---|-------------------|
| T | F                 |

| $p$ | $p \wedge \neg p$ |
|-----|-------------------|
| F   | F                 |

This is called the Law of Contradiction.

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## Contingency

A proposition that is sometimes true and sometimes false.

Example:

$$p \wedge q$$

Its truth depends on the truth values of  $p$  and  $q$ .

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## 2.9 Laws of Logical Equivalence

These laws are used to simplify logical expressions.

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### Identity Laws

$$p \wedge T \equiv p$$

$$p \vee F \equiv p$$

Example:

$$(p \wedge T) \vee q$$

$$\equiv p \vee q$$


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### Domination Laws

$$p \vee T \equiv T$$

$$p \wedge F \equiv F$$

Example:

$$(x > 5) \vee T$$

is always true.

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## Idempotent Laws

$$p \vee p \equiv p$$

$$p \wedge p \equiv p$$

Example:

$$(x > 0) \vee (x > 0)$$

$$\equiv x > 0$$

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## Double Negation Law

$$\neg(\neg p) \equiv p$$

Example:

"It is not true that it is not raining"

means

"It is raining."

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## Commutative Laws

$$p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

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## Associative Laws

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

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## Distributive Laws

$$p \wedge (q \vee r)$$

$\equiv$

$$(p \wedge q) \vee (p \wedge r)$$

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## Absorption Laws

$$p \vee (p \wedge q)$$

$\equiv p$

$$p \wedge (p \vee q)$$

$\equiv p$

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## De Morgan's Laws

First Law:

$$\neg(p \wedge q)$$

$\equiv$

$$\neg p \vee \neg q$$

Second Law:

$$\neg(p \vee q)$$

$\equiv$

$$\neg p \wedge \neg q$$

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## Example of De Morgan's Law

Statement:

"It is not true that Ram and Shyam both passed."

Equivalent Statement:

"Ram failed or Shyam failed."

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## Summary

In this chapter we studied:

- Conditional Statements
- Truth Table of Implication
- Converse
- Inverse
- Contrapositive
- Biconditional Statements
- Necessary and Sufficient Conditions
- Logical Equivalence
- Tautologies
- Contradictions
- Contingencies
- Laws of Logical Equivalence
- De Morgan's Laws

These concepts provide the foundation for Predicate Logic, Proof Techniques, and Mathematical Reasoning.